

# Manual Bidding of Balancing- and Transport Power

## Amendments to Register:

| Version number | Date              | Amendment                                                                                                                                                  |
|----------------|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0.1            | 18 September 2000 | Initial version                                                                                                                                            |
| 0.2            | 27 September 2000 | Incorporation of project members' comments<br>Incorporation of new version of MIG QUOTES 0.5                                                               |
| 0.3            | 20 October 2000   | - Workshop version<br>- Incorporation of definitive version of MIG QUOTES 1.0                                                                              |
| 1.0            | 04 December 2000  | - Definitive version<br>- Adjustment of permissible values for object of regulation and dispatch time<br>- Amendment of closing term as at day of delivery |
| 1.1            | 12 December 2001  | Revision of objects of regulation and update of handbook                                                                                                   |
| 1.2            | 12 May 2003       | Changes to chapter 2 (PTU adjustment)                                                                                                                      |
| 2.0            | June 2010         | Adaptation to UTIL TS, Blockbids                                                                                                                           |
| 2.1            | February 2011     | Adjustment limits Blockbids                                                                                                                                |
| 2.2            | 25 May 2012       | Adjustment Gate Closure Time                                                                                                                               |
| 2.3            | November 2017     | Adjustment Power                                                                                                                                           |
| 2.4            | January 2018      | Adjustment Gate Closure Time                                                                                                                               |
| 3.0            | January 2019      | Update lay-out,<br>Adjusted content in accordance with the amendment of 'Netcode Elektriciteit' (ACM/18/032994)                                            |
| 4.0            | January 2020      | Obligation contract number aFRR changed to optional                                                                                                        |
| 4.1            | May 2021          | Removal freeze period day of preparation                                                                                                                   |
| 5              | 16 March 2022     | Removal mFRRsa<br>Change Ramping rate                                                                                                                      |
| 5.1            | 5 July 2024       | Update location field                                                                                                                                      |
| 5.2            | 18 May 2026       | Update ramp rate 200%                                                                                                                                      |

## 1. Introduction

This document contains instructions for submitting bids balancing- and transport power (BTP)<sup>1</sup> to TenneT TSO, as guideline for BSPs<sup>2</sup> and CSPs<sup>3</sup>. Topics that will be discussed in this document are the attributes of the bid (significance, permissible values), the timetable and the transitions from summer to winter time.

Further information on balancing and transport power can be found in the 'Netcode Elektriciteit', the Implementation Rules and the product information documents on [www.tennet.eu](http://www.tennet.eu). In case of differences between the texts, the text of the 'Netcode Elektriciteit' prevails. If the topic is not described in the 'Netcode Elektriciteit' the text of the Implementation Rules shall prevail.

BTV bids are options, whereby TenneT is obligated, after activation, to:

- calculate a certain volume based on the characteristics of bidding and activation
- settle this volume with the BSP or CSP on at least its financial terms.

adjust this volume on the imbalance of the BRP behind the activated allocation points

The BSP/CSP is responsible for meeting the bidding requirements; incorrect bids will be rejected by TenneT. TenneT is responsible for timely activation, and respecting the bidders minimum requirements with respect to the preparation period<sup>4</sup> and the delivery period<sup>5</sup>.

TenneT is responsible for correct settlement of activated bids and ensuing adjustment of imbalances. The BSP/CSP is responsible for the delivery of the energy, corresponding to the bid activated by TenneT, during the delivery period.

Note that any reference to a time of day shall be CET.

### 1.1 Categories bids balancing- and transport power

All bids BTP belong to one of the following:

1. Balancing purposes:
  - a. Bids aFRR<sup>6</sup>, preparation period = 0, delivery period = 1
2. Other purposes
  - a. Bids Reserve power, preparation period  $\geq 3$ , delivery period  $\geq 4$

As the requirements for bid messages are similar for aFRR and Reserve power other purposes, this manual will use the generic term BTP.

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<sup>1</sup> BTP includes aFRR, mFRRs and Reserve power other purposes.

<sup>2</sup> BSP: Balancing Service Provider

<sup>3</sup> CSP; Transport Service Provider

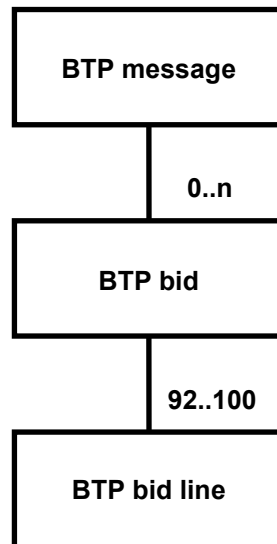
<sup>4</sup> Preparation period was previously known as Activation time

<sup>5</sup> Delivery period was previously known as Activation duration

<sup>6</sup> Automatic Frequency Restoration Reserve; previously known as regulating power

## 2. Structure of BTP message

The diagram below shows the structure of the messages of BTP submitted by the BSP/CSP:



The BTP message is made up of a combination of BTP bids, whose number per BTP message is unlimited (n).

A BTP message with number of BTP bids nil (0) indicates the BSP/CSP does not want to submit BTP. Each new BTP message sent overrides all previous BTP messages.

The BTP messages are exchanged in a standardised format, in compliance with the EDI standard for the Dutch energy sector (EDINE).

A detailed description of this format is available at MijnEDSN at [www.edsn.nl](http://www.edsn.nl)

The BTP message is sent to the CPS, the central mailbox system, and should be addressed to TenneT/SO.

Each bid BTP is composed of bidding lines, one for each ISP<sup>7</sup> (clock quarter of an hour).

Any standard day counts 96 ISPs. On days with transitions to and from daylight savings time there are 92 or 100 bidding lines. (See also chapter 4)

Number of positions and character (numeric, alphanumeric, ...) of the attributes are defined in the UN/Cefact UNSM (in this case UTILTS). An exception is Data Element C506.1154 in the RFF-segment: the number of positions for this Data Element in the messages defined in this document is an..35 instead of the standard

an..70 specified in the UNSM.

## 2.1 Attributes of BTP message

The BSP/CSP is required to confer values upon the following attributes in its BTP message:

| <b>Attribute</b> | <b>Unit</b> | <b>Description</b>                                                                                                                                                                                                 | <b>Permissible values</b>                                                                                                              |
|------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| BSP/CSP          | N/A         | Identification of the supplier of balancing- and/or transport power                                                                                                                                                | EAN code                                                                                                                               |
| BRP              | N/A         | Identification of the BRP (Balance Responsible Party) whose imbalance will be adjusted upon activation of transport power.<br><br>In case of aFRR the BRP information is send via an additional information stream | EAN code<br><br>aFRR: This has to be the EAN-code of the BSP itself. The BRP information is send via an additional information stream. |
| Request          | N/A         | If the message is submitted at TenneT's request, the TenneT-issued request number must be included                                                                                                                 | TenneT-issued request number                                                                                                           |
| Date of delivery | N/A         | The date for which the bids relate                                                                                                                                                                                 | Date in the range <sup>8</sup> current and current + 7 days                                                                            |

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<sup>8</sup> Inclusive threshold values

## 2.2 Attributes of BTP bid

Each bid in the BTP message is specified through the following attributes:

| <b>Attribute</b>     | <b>Unit</b>  | <b>Description</b>                                                                                                                                                                                                                          | <b>Permissible values</b>                                                                                                      |
|----------------------|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Contract             | N/A          | Identification of the contract between the BSP/CSP and TenneT                                                                                                                                                                               | TenneT-issued contract number comprising 10 alphanumeric characters                                                            |
| Reference            | N/A          | BSP/CSP-issued unique identification of the bid as part of the message                                                                                                                                                                      | BSP/CSPs choice                                                                                                                |
| Object               | N/A          | An object enables a BSP/CSP to couple two bids. From an Object only one bid can be activated. See notes below the table.                                                                                                                    | BSP/CSPs choice                                                                                                                |
| Preparation period   | ISP          | ISP interval relative to current for which Bid is available to be activated by TenneT; distinguishes aFRR and Reserve power other purposes                                                                                                  | Integer value 0 for aFRR or in the range <sup>8</sup> 3 to 672 (7 days) for Reserve power other purposes                       |
| Delivery period      | ISP          | Minimum number of consecutive ISPs for admissible activation by TenneT; distinguishes aFRR and Reserve power other purposes                                                                                                                 | Integer value 1 for aFRR or in the range <sup>8</sup> 4 to 672 range (7 days) for Reserve power other purposes                 |
| Power                | MW           | Quantity<br>+ upward<br>- downward                                                                                                                                                                                                          | Upward: Integer in range <sup>8</sup> 1 <sup>9</sup> to 999<br>Downward: Integer in range <sup>8</sup> -1 <sup>9</sup> to -999 |
| Ramping rate         | % per minute | Ramping rate, as percentage of bid quantity per minute                                                                                                                                                                                      | One decimal place, value in the range <sup>8</sup> 20.0 to 200.0                                                               |
| Location/Grid object | N/A          | A connection, or set of connections, within the Dutch high-voltage grid, from which bidder will dispatch on activation. This connection or set of connections belongs to one owner or administrator. For aFRR it refers to the LFC area NL. | EAN code                                                                                                                       |

Note:

- A aFRR object couples two bids of opposite sign (upward/downward)
- A Reserve power object other purposes couples two bids with similar quantity and preparation period, but with different delivery period and price.

<sup>9</sup> A message is only allowed to contain 3 bids with a size smaller than 4 MW

## 2.3 Attributes of BTP bid line

Each ISP on the date of delivery for which the bid is available must be specified.

| <b>Attribute</b> | <b>Unit</b> | <b>Description</b>               | <b>Permissible values</b>                                                        |
|------------------|-------------|----------------------------------|----------------------------------------------------------------------------------|
| Availability     | ISP         | ISP number for which bid applies | Unique Integer value in range <sup>8</sup> 1 to 100, ascending                   |
| Bid price        | €/MWh       | Energy price                     | Two decimal places<br>Value in the range <sup>8</sup> –100,000.00 to +100,000.00 |

Note:

- For bids FRR the bid price may vary per ISP; for bids for other purposes the bid price must be constant.
- The sign of the product of the quantity and the settlement price denotes the direction of the cash flow, with + indicating that that pays the BSP/CSP and -, that the BSP/CSP that pays TenneT.

|                      | Bid price > 0       | Deployment Bid price < 0 |
|----------------------|---------------------|--------------------------|
| Power > 0 (upward)   | TenneT pays BSP/CSP | BSP/CSP pays TenneT      |
| Power < 0 (downward) | BSP/CSP pays TenneT | TenneT pays BSP/CSP      |

- € 1.23/MWh = € 0.00123/kWh

## 2.4 Optional and mandatory aspects of attributes

As the table below shows, the attributes of the message, the bid and the bidding line are either mandatory or optional depending on the category of BTP bid:

| <b>Attribute</b>   | <b>aFRR</b> | <b>Reserve power<br/>other purposes</b> |
|--------------------|-------------|-----------------------------------------|
| BSP/CSP            | Mandatory   | Mandatory                               |
| BRP                | Mandatory   | Mandatory                               |
| Request            | Optional    | Optional                                |
| Date of delivery   | Mandatory   | Mandatory                               |
| Contract           | N/A         | N/A                                     |
| ID                 | Mandatory   | Mandatory                               |
| Object             | Optional    | Optional                                |
| Preparation period | Mandatory   | Mandatory                               |
| Delivery period    | Mandatory   | Mandatory                               |
| Ramping rate       | Mandatory   | N/A                                     |
| Location           | Mandatory   | Mandatory                               |
| Power              | Mandatory   | Mandatory                               |
| Bid price          | Mandatory   | Mandatory                               |

Note:

- Mandatory implies that a value must be specified, in accordance with admissible or prescribed values in 2.2, 2.3 and 2.4
- Optional implies that a value may be specified, in accordance with admissible values in 2.2, 2.3 and 2.4
- N/A implies that no value is allowed to be specified.

### 3. Timetable

The BTP messages are required to be sent in accordance with a fixed timetable so as to enable TenneT's timely processing of the bids. Time of receipt in the CPS (central mailbox system) is deciding.

#### 3.1 Day of preparation (D-1)

The BTP bids for which there is a capacity contract for the day of delivery are required to reach TenneT by 14.45 CET daily on the day prior to that of delivery.

#### 3.2 Day of delivery

On the day of delivery, the deadline for submitting revised BTP bids always closes 30 minutes ahead of each ISP.

Example:

It is 19.12 CET and a BSP/CSP decides to revise its BTP bids for the current day. At this time the BSP/CSP will be permitted to submit changes for the ISP from 19.45 to 20.00 CET and all further ISPs. In the event of the BSP/CSP having changed one or more ISPs preceding that from 19.45 to 20.00 CET, all its BTP bids will be rejected.

### 4. Transition to and from daylight savings time

On the day of transition from summer (i.e. daylight saving) to winter time (clock put back at 3.00 a.m.), four additional bidding lines are required to be provided in respect to the third hour. This implies for those BSP/CSPs which compile their bids using Excel templates that all bidding lines relating to the fourth and further hours are put back four rows.

A similar mechanism applies on the day of transition from winter to summer time (clock put forward at 2.00 a.m.). Here there are no bidding lines for the third hour, which implies that all bidding lines relating to the fourth and further hours are moved up four lines.



In the following examples the periods that deviate from a 'normal day' are marked.

Transition from summer to  
winter time  
(100 bidding lines)

| Period<br>(Excel) | Clock hour    |
|-------------------|---------------|
| ....              | ....          |
| 01:30 - 01:45     | 01:30 - 01:45 |
| 01:45 - 02:00     | 01:45 - 02:00 |
| 02:00 - 02:15     | 02:00 - 02:15 |
| 02:15 - 02:30     | 02:15 - 02:30 |
| 02:30 - 02:45     | 02:30 - 02:45 |
| 02:45 - 03:00     | 02:45 - 03:00 |
| 03:00 - 03:15     | 02:00 - 02:15 |
| 03:15 - 03:30     | 02:15 - 02:30 |
| 03:30 - 03:45     | 02:30 - 02:45 |
| 03:45 - 04:00     | 02:45 - 03:00 |
| 04:00 - 04:15     | 03:00 - 03:15 |
| ....              | ....          |
| ....              | ....          |
| 24:00 - 24:15     | 23:00 - 23:15 |
| 24:15 - 24:30     | 23:15 - 23:30 |
| 24:30 - 24:45     | 23:30 - 23:45 |
| 24:45 - 25:00     | 23:45 - 24:00 |

Transition from winter to  
summer time  
(92 bidding lines)

| Period<br>(Excel) | Clock hour    |
|-------------------|---------------|
| ....              | ....          |
| 01:30 - 01:45     | 01:30 - 01:45 |
| 01:45 - 02:00     | 01:45 - 02:00 |
| 02:00 - 02:15     | 03:00 - 03:15 |
| 02:15 - 02:30     | 03:15 - 03:30 |
| ....              | ....          |
| ....              | ....          |
| 22:15 - 22:30     | 23:15 - 23:30 |
| 22:30 - 22:45     | 23:30 - 23:45 |
| 22:45 - 23:00     | 23:45 - 24:00 |
| 23:00 - 23:15     | N/A           |
| 23:15 - 23:30     | N/A           |
| 23:30 - 23:35     | N/A           |
| 23:45 - 24:00     | N/A           |
| 24:00 - 24:15     | N/A           |
| 24:15 - 24:30     | N/A           |
| 24:30 - 24:45     | N/A           |
| 24:45 - 25:00     | N/A           |